Minerals and Energy Resources

The big idea

Almost everything we use, from a pin to a train, is made from **minerals**. And every factory, vehicle and home needs **energy**. Minerals are **non-renewable** and unevenly distributed, and most of our energy comes from **fossil fuels** that will run out. This chapter covers how minerals occur, India's major minerals and their belts, conventional and non-conventional energy, and why conservation matters.

MINERAL

Geologists define a mineral as a **"homogenous, naturally occurring substance with a definable internal structure."** Minerals are found in varied forms, from the hardest diamond to the softest talc. **Rocks** are combinations of homogenous substances called minerals.

Mode of occurrence of minerals

Minerals are usually found in **'ores'** (an accumulation of any mineral mixed with other elements). The mineral content of the ore must be in sufficient concentration to make extraction commercially viable.

1. **Igneous and metamorphic rocks**: minerals occur in **cracks, crevices, faults or joints**. Smaller occurrences are called **veins** and larger ones **lodes**. They are formed when minerals in liquid/molten and gaseous forms are forced upward through cavities. Examples: **tin, copper, zinc and lead**.
2. **Sedimentary rocks**: minerals occur in **beds or layers**, formed by deposition, accumulation and concentration in horizontal strata. **Coal** and some forms of **iron ore** are concentrated by long periods under great heat and pressure;

others like **gypsum, potash salt and sodium salt** form by evaporation, especially in arid regions.

3. **Decomposition of surface rocks**: soluble constituents are removed, leaving a **residual mass** of weathered material containing ores. **Bauxite** is formed this way.
4. **Alluvial deposits** in the sands of valley floors and the base of hills: called **'placer deposits'**, they generally contain minerals not corroded by water: **gold, silver, tin and platinum**.
5. **Ocean waters**: most minerals in ocean water are too widely diffused to be of economic significance, but **common salt, magnesium and bromine** are largely derived from ocean waters. Ocean beds are rich in **manganese nodules**.

RAT-HOLE MINING

In **Jowai and Cherrapunjee (Meghalaya)**, coal mining is done by family members in the form of a long narrow tunnel known as **'rat hole' mining**. In most of India minerals are a nationalised property, but in tribal north-east India they are owned by individuals or communities.

Peninsular rocks contain most reserves of coal, metallic minerals, mica and many non-metallic minerals; **sedimentary rocks** on the western and eastern flanks of the peninsula (Gujarat and Assam) have most petroleum deposits; **Rajasthan** with its rock systems has reserves of many non-ferrous minerals; the vast alluvial plains of north India are almost devoid of economic minerals.

Ferrous minerals

Ferrous minerals account for about **three-fourths** of the total value of metallic mineral production.

IRON ORE

The basic mineral and the backbone of industrial development. India is rich in good-quality iron ores.

- **Magnetite**: the **finest** iron ore with a very high content of iron, **up to 70%**. Has excellent **magnetic** qualities, valuable for the electrical industry.
- **Hematite**: the **most important industrial iron ore** in terms of quantity used, but slightly lower iron content than magnetite (**50–60%**).
  Major iron ore belts:

1. **Odisha–Jharkhand belt**: high-grade hematite at **Badampahar mines** in Mayurbhanj and Kendujhar districts of Odisha; in **Gua and Noamundi** in Singhbhum district of Jharkhand.
2. **Durg–Bastar–Chandrapur belt** (Chhattisgarh and Maharashtra): very high-grade hematites in the famous **Bailadila range of hills** in Bastar district, Chhattisgarh; the range has 14 deposits of super high-grade hematite with the best physical properties for steel making. Iron ore from these mines is **exported to Japan and South Korea via Visakhapatnam port**.
3. **Ballari–Chitradurga–Chikkamagaluru–Tumakuru belt** (Karnataka): large reserves. The **Kudremukh mines** in the Western Ghats of Karnataka are a **100 per cent export unit**; Kudremukh deposits are known to be one of the largest in the world. The ore is transported as **slurry through a pipeline** to a port near **Mangaluru**.
4. **Maharashtra–Goa belt**: includes **Goa** and **Ratnagiri** district of Maharashtra. Ores are not of very high quality but are efficiently exploited; exported through **Marmagao port**.

Manganese is mainly used in the manufacturing of **steel and ferro-manganese alloy**. Nearly **10 kg of manganese** is required to manufacture **one tonne of steel**. Also used for bleaching powder, insecticides and paints. **Odisha** is the largest producer of manganese ores in India.

Non-ferrous minerals

India's reserves and production of non-ferrous minerals are not very satisfactory.

- **Copper**: India is critically deficient in copper reserves. Being malleable, ductile and a good conductor, copper is mainly used in electrical cables, electronics and chemical industries. The **Balaghat mines** in Madhya Pradesh, the **Khetri mines** in Rajasthan and the **Singhbhum district** of Jharkhand are leading producers.
- **Bauxite**: a clay-like substance from which **alumina and later aluminium** is obtained. Bauxite deposits form by the decomposition of rocks rich in **aluminium silicates**. Aluminium combines strength of metals like iron with extreme lightness, good conductivity and great malleability. Deposits are mainly found in the **Amarkantak plateau, Maikal hills and the plateau region of Bilaspur-Katni**. **Odisha** was the largest bauxite producing state in India; **Panchpatmali deposits** in **Koraput district** are the most important.

Non-metallic minerals

- **Mica**: made up of a series of plates or leaves; splits easily into thin sheets. Can be clear, black, green, red, yellow or brown. Due to excellent **di-electric strength**, low power loss factor, insulating properties and resistance to high voltage, it is one of the most indispensable minerals used in **electric and electronic industries**. Deposits are found on the **northern edge of the Chota Nagpur plateau**. The **Koderma–Gaya–Hazaribagh belt of Jharkhand** is the leading producer. In Rajasthan, the major producing area is around **Ajmer**; **Nellore** mica belt of Andhra Pradesh is also important.
- **Limestone** is found in association with rocks composed of calcium carbonates or calcium and magnesium carbonates. It is the basic raw material for the **cement industry** and essential for **smelting iron ore** in the blast furnace.

Hazards of mining: dust and noxious fumes inhaled by miners make them vulnerable to **pulmonary diseases**; risk of collapsing mine roofs, inundation and fires; water sources in the region get contaminated; dumping of waste and slurry leads to degradation of land, soil and increased stream and river pollution.

Conservation of minerals: geological processes of mineral formation are so slow that the rates of replenishment are infinitely small compared to present consumption. Minerals are **finite and non-renewable**. We should use them in a planned and sustainable manner, use improved technologies to use low-grade ores at low costs, **recycle metals** (using scrap metals), and use **substitutes**.

Energy resources: conventional sources

Energy can be generated from fuel minerals like coal, petroleum, natural gas, uranium and from electricity. **Conventional sources**: firewood, cattle dung cake, coal, petroleum, natural gas and electricity (hydel and thermal). **Non-conventional**: solar, wind, tidal, geothermal, biogas and atomic energy. In rural India, firewood and cattle dung cake are most common; over 70% of rural household energy comes from these. But continuing them is difficult because forest area is decreasing and dung is better used as manure.

COAL

The most abundantly available fossil fuel in India; provides a substantial part of the nation's energy needs. Formed due to the compression of plant material over millions of years. Depending on the degree of compression, depth and time of burial:

- **Peat**: decaying plants in swamps; low carbon, high moisture, low heating capacity.
- **Lignite**: low-grade **brown coal**, soft with high moisture content. Principal reserves at **Neyveli (Tamil Nadu)**; used for generating electricity.
- **Bituminous**: coal buried deep and subjected to increased temperatures; the most popular coal in commercial use. **Metallurgical coal** is high-grade bituminous coal of special value for smelting iron in blast furnaces.
- **Anthracite**: the **highest quality hard coal**.
  Two main groups in India: Gondwana coal, a little over **200 million years** old, and **Tertiary** deposits, only about **55 million years** old. Major Gondwana resources are in the **Damodar valley (West Bengal–**

Jharkhand): **Jharia, Raniganj and Bokaro** are important coalfields; the **Godavari, Mahanadi, Son and Wardha** valleys also contain coal. **Tertiary coals** occur in the north-eastern states of **Meghalaya, Assam, Arunachal Pradesh and Nagaland**. Coal is a bulky material that loses weight on use, so heavy industries and thermal power stations are located on or near coalfields.

PETROLEUM

Petroleum (mineral oil) is the next major energy source after coal. It provides fuel for heat and lighting, lubricants for machinery and raw materials for many manufacturing industries. Petroleum refineries act as a **"nodal industry"** for synthetic textile, fertiliser and numerous chemical industries. Most petroleum occurrences in India are associated with **anticlines and fault traps** in rock formations of the tertiary age. In regions of folding, anticlines or domes, oil is trapped in the crest of the upfold; gas, being lighter, usually occurs above the oil.

- About **63 per cent** of India's petroleum production is from **Mumbai High**, **18 per cent** from **Gujarat** and **16 per cent** from **Assam**.
- **Ankleshwar** is the most important field of Gujarat.
- **Assam** is the **oldest** oil-producing state of India. **Digboi, Naharkatiya and Moran-Hugrijan** are important oil fields there.

NATURAL GAS

An important clean energy resource found in association with or without petroleum. Used as a source of energy as well as an industrial raw material in the **petrochemical** industry; considered an environment-friendly fuel because of low CO_2 emissions. Large reserves have been discovered in the **Krishna-Godavari basin**; along the west coast reserves are found in the **Mumbai High** and allied fields, supplemented by finds in the **Gulf of Khambhat**; the **Andaman and Nicobar Islands** also have large reserves. The **1,700 km long Hazira-Vijaipur-Jagdishpur (HVJ)** cross-country gas pipeline links Mumbai High and Bassien with fertiliser, power and industrial

complexes in western and northern India. **CNG** is used as a liquid fuel substitute for vehicles.

Electricity is generated mainly in two ways:

- **Hydro electricity**: by **fast-flowing water**, a renewable resource. Multi-purpose projects like **Bhakra Nangal, Damodar Valley Corporation and the Kopili Hydel Project** produce hydroelectric power.
- **Thermal electricity**: by burning **coal, petroleum and natural gas** to drive turbines. Thermal power stations use non-renewable fossil fuels.

Non-conventional sources of energy

Growing consumption of energy has made India increasingly dependent on fossil fuels. Rising prices of oil and gas and their potential shortages have raised uncertainties about future energy security, and fossil fuels cause pollution. So there is a pressing need to use renewable sources.

- **Nuclear or atomic energy**: obtained by **altering the structure of atoms**; enormous energy is released as heat and used to generate electricity. **Uranium and thorium**, available in **Jharkhand and the Aravalli ranges of Rajasthan**, are used. The **monazite sands of Kerala** are also rich in **thorium**. Nuclear power stations: **Narora (UP), Rawatbhata (Rajasthan), Kakrapar (Gujarat), Tarapur (Maharashtra), Kaiga (Karnataka), Kalpakkam (Tamil Nadu)**.
- **Solar energy**: India is a tropical country with enormous possibilities of tapping solar energy. **Photovoltaic technology** converts sunlight directly into electricity. The largest solar plant of India at the time was located at **Madhapur, near Bhuj (Gujarat)**, where solar energy is used to **sterilise milk cans**. Solar energy can reduce rural dependence on firewood and dung cakes.
- **Wind power**: India ranks among the "**Wind Super Power**" countries of the world. The largest wind farm cluster is located in **Tamil Nadu from Nagercoil to Madurai**. Andhra Pradesh, Karnataka, Gujarat, Kerala, Maharashtra and

Lakshadweep have important wind farms; **Nagercoil and Jaisalmer** are well known for effective use of wind energy.

- **Biogas**: shrubs, farm waste, animal and human waste are used to produce biogas for domestic consumption in rural areas. Decomposition of organic matter yields gas with higher thermal efficiency than kerosene, dung cake and charcoal. Plants using cattle dung are known as **'Gobar gas plants'**; they provide energy and improved quality manure, and prevent loss of trees and manure due to burning of fuelwood and cow dung cakes.
- **Tidal energy**: oceanic tides can be used to generate electricity. Floodgate dams are built across inlets; during high tide water flows into the inlet and gets trapped when the gate is closed; it then flows back through a pipe that carries it through a power-generating turbine. In India, the **Gulf of Khambhat, the Gulf of Kuchchh** in Gujarat on the western coast and the **Gangetic delta in the Sunderban regions** of West Bengal provide ideal conditions.
- **Geothermal energy**: heat and electricity produced by using the **heat from the interior of the Earth**. Where the geothermal gradient is high, high temperatures are found at shallow depths; groundwater absorbs heat and rises as hot springs. Two experimental projects: one in the **Parvati valley near Manikarn (Himachal Pradesh)** and the other in the **Puga valley, Ladakh**.

Conservation of energy resources

Energy is a basic requirement for economic development. We need a cautious approach for judicious use of limited energy resources: use **public transport** instead of individual vehicles, switch off electricity when not in use, use **power-saving devices**, and use **non-conventional sources** of energy. "**Energy saved is energy produced.**"

MAGNETITE VS HEMATITE

Magnetite is the **finest** ore (up to 70% iron); hematite is the **most important industrial** ore by quantity (50–60%). Don't call hematite the richest.

BOARD FAVOURITES AND MAP WORK

"Modes of occurrence of minerals", "Distinguish ferrous/non-ferrous, conventional/non-conventional", "Why do we need to conserve minerals?", "Coal types", "Nuclear/solar/wind energy in India". **Map: Iron ore mines**: Mayurbhanj, Durg, Bailadila, Ballari, Kudremukh. **Coal mines**: Raniganj, Bokaro, Talcher, Neyveli. **Oil fields**: Digboi, Naharkatiya, Mumbai High, Bassien, Kalol, Ankleshwar. **Thermal power plants**: Namrup, Singrauli, Ramagundam. **Nuclear power plants**: Narora, Kakrapar, Tarapur, Kalpakkam.

Quick recap

- Minerals occur in veins/lodes, beds/layers, residual mass, placer deposits and ocean water.
- Iron ore: magnetite (finest), hematite (most used); four belts incl. Bailadila and Kudremukh.
- Odisha leads manganese and bauxite (Panchpatmali, Koraput); copper at Balaghat, Khetri, Singhbhum; mica in Koderma–Gaya–Hazaribagh.
- Coal: Gondwana (200 million yrs, Damodar valley) and Tertiary (55 million yrs, north-east); lignite at Neyveli.
- Petroleum: Mumbai High 63%, Gujarat 18%, Assam 16%; HVJ pipeline 1,700 km.
- Non-conventional: nuclear, solar (Madhapur), wind (Nagercoil–Madurai), biogas, tidal (Khambhat), geothermal (Manikarn, Puga).

Practice questions

1. Placer deposits found in the sands of valley floors usually contain
   (a) coal and petroleum (b) bauxite and mica (c) gold, silver, tin and platinum (d) gypsum and potash salt
2. Which is the finest iron ore with up to 70 per cent iron?
   (a) Magnetite (b) Hematite (c) Limonite (d) Siderite
3. The Kudremukh iron ore mines are located in
   (a) Odisha (b) Chhattisgarh (c) Goa (d) Karnataka
4. Rat-hole mining is practised in
   (a) Jharia (b) Jowai and Cherrapunjee, Meghalaya (c) Neyveli (d) Bailadila
5. Which state is the oldest oil-producing state of India?
   (a) Gujarat (b) Maharashtra (c) Assam (d) Rajasthan
6. Lignite, a low-grade brown coal, is mainly found at
   (a) Raniganj (b) Jharia (c) Bokaro (d) Neyveli
7. Geothermal energy projects in India are located at
   (a) Bhuj and Jaisalmer (b) Manikarn (Parvati valley) and Puga valley (c) Narora and Kaiga (d) Gulf of Khambhat and Sunderbans
8. Distinguish between conventional and non-conventional sources of energy.
9. Why do we need to conserve mineral resources? Suggest ways to do so.

Answers: 1. c 2. a 3. d 4. b 5. c 6. d 7. b 8. see notes 9. see notes